Rings of Presence: one balance rule for gravitational boundaries, and the galactic acceleration scale as $c^2\sqrt{\Lambda/3}\,/\,2\pi$

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October 2026

Abstract

We describe weak-field gravity with one dimensionless field, the *presence* $P = \sum_i GE_i/(c^4 r_i) = -\Phi/c^2$, whose value, gradient and Hessian give clock rates, accelerations and tides. This is a reorganisation of weak-field general relativity, not a modification, and it reproduces the standard tests by construction. We use it for two results. (i) The Hill radius, the Jacobi tidal radius, the L1 distance and the largest turnaround radius allowed by a cosmological constant all follow from one balance condition, $r^3 = Gm/S$, where S is the stretching rate of the environment, ending in $S = \Lambda c^2/3$ for empty space. The outermost moons of six planets lie within $0.46\,r$, 107 of 111 satellites of the Milky Way and M31 lie inside their hosts' radii, and the zero-velocity radii of the Local Group and Virgo lie inside their Λ limits. (ii) Treating empty space as the outermost host of every system, we propose that the acceleration scale of the radial acceleration relation is $a_0 = a_\Lambda/k$, with $a_\Lambda = c^2\sqrt{\Lambda/3}$ and k the total presence of a point source inside a ring divided by its presence at the edge, in units of the ring's size. This integral equals 2π for both planar and spherical rings, which gives a reason for the long-noted coincidence $2\pi a_0 \approx c^2(\Lambda/3)^{1/2}$ and a parameter-free prediction, $a_0 = 0.86 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ (Planck) or $0.94 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ (SH0ES). Sixteen gas-rich SPARC galaxies give $0.82^{+0.22}_{-0.17} \times 10^{-10}\,\mathrm{m\,s^{-2}}$, in agreement. The full sample of 164 galaxies gives $1.14 \pm 0.08 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ at the fiducial stellar mass-to-light ratio $\Upsilon_{\rm disk} = 0.5$ and matches the prediction for $\Upsilon_{\rm disk} \approx 0.6$. Per-galaxy a_0 shows no dependence on size, distance, surface brightness, rotation speed, gas fraction or bulge fraction, as the scale-free integral requires. Forty-one disks at $z = 0.65$–2.45 show no rise of a_0 with redshift, though they cannot yet exclude $a_0 \propto H(z)$. We state which observations can confirm or exclude the proposal.

1 Introduction

Rotationally supported galaxies obey a tight radial acceleration relation (RAR) between the observed centripetal acceleration $g_{\rm obs}$ and the acceleration $g_{\rm bar}$ produced by their baryons (McGaugh et al., 2016; Lelli et al., 2017). The relation is Newtonian at high acceleration and departs from it below a scale $a_0 \approx 1.2 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ that appears to be the same in every galaxy (Li et al., 2018). Whether the RAR reflects modified dynamics (Milgrom, 1983; Famaey & McGaugh, 2012) or emerges from galaxy formation with cold dark matter (Keller & Wadsley, 2017; Ludlow et al., 2017), the numerical value of a_0 is an empirical input. It has long been noted that $2\pi a_0 \approx cH_0 \approx c^2(\Lambda/3)^{1/2}$ (Milgrom, 1983, 1999, 2016), but the factor 2π has no accepted derivation.

A second link between local dynamics and Λ is well understood: a cosmological constant sets a largest possible turnaround radius, $R_\Lambda = (3GM/\Lambda c^2)^{1/3}$, for any bound structure (Pavlidou & Tomaras, 2014).

This paper organises weak-field gravity around one scalar, the *presence*, and uses it for two results. We are explicit about which parts are new. Sections 2 and 3 restate known physics in

one language: the presence is the Newtonian potential divided by $-c^2$, and the boundary rule unifies results that are individually standard. Their value is that one number describes every scale, from a moon to the edge of a galaxy group, and that empty space appears as the outermost environment of every system. Section 4 contains the one proposal that goes beyond current physics: a derivation of the factor 2π in the a_0–Λ relation from a presence integral, with no adjustable parameter. We test it on the SPARC rotation curves (Lelli et al., 2016) and on disks at $z \approx 1$–2.4 (Price et al., 2021). Section 5 discusses the stellar mass-to-light systematic, which currently limits the test, and the observations that can decide. Every number in the paper is produced by public scripts.

2 Presence: weak-field gravity as one scalar

For sources with energies E_i (for matter at rest $E_i = m_i c^2$) at distances r_i,

$$P(\mathbf{x}) = \sum_i \frac{G E_i}{c^4\, r_i} = -\frac{\Phi(\mathbf{x})}{c^2}, \tag{1}$$

where Φ is the Newtonian potential. P is dimensionless and small: about 7×10^{-10} at Earth's surface, 10^{-8} from the Sun at 1 au and 2×10^{-6} at the solar surface. In the isotropic weak-field metric $ds^2 = -(1-2P)\,c^2 dt^2 + (1+2P)\,d\mathbf{x}^2$, three readings of the same field carry most of weak-field gravity:

- its *value* sets clock rates, $d\tau/dt \simeq 1 - P - v^2/2c^2$;
- its *gradient* is the acceleration, $\mathbf{g} = c^2 \nabla P$;
- its *Hessian* is the tidal tensor, $T_{ij} = c^2\, \partial_i \partial_j P$, of magnitude $2GMd/D^3$ across a body of size d at distance D from a mass M.

Thought experiment: the room inside a shell. Place a laboratory at the centre of a large, uniform shell of matter. The shell raises P everywhere inside by the same amount, so every clock in the room slows equally and nothing is pulled anywhere. No local experiment can detect the shell. Only differences, slopes and changes of slope of P are observable. This is why the presence of the whole Universe, of order unity at every point, has no local effect.

Light deflection, perihelion advance and radar delay follow from the metric as in general relativity, and orbital decay by gravitational radiation from the Peters–Mathews luminosity. Table 1 lists the values we recover. They agree because the description is equivalent to weak-field general relativity; we list them only to show that the bookkeeping is right.

3 One balance rule for gravitational boundaries

3.1 The rule

A body of mass m sits in an environment whose tidal field, in the frame moving with the body, stretches nearby test particles at a rate S. Here S is the largest eigenvalue of the effective tidal tensor, including the centrifugal term when the frame rotates. Along that direction a test particle at distance r feels $-Gm/r^2 + S\,r$. The body controls the region where its own pull wins, which ends at

$$r_{\rm own}^3 = \frac{G\,m}{S}. \tag{2}$$

The environment fixes S:

- **Point-mass host** M at distance D, circular orbit: $S = \Omega^2 - \partial^2\Phi/\partial D^2 = 3GM/D^3$. This gives $r_{\rm own} = D\,(m/3M)^{1/3}$, the Hill radius and the distance to L1 and L2 (Murray & Dermott, 1999).

Table 1: Standard tests recovered by the presence description (identical to general relativity by construction). Measurements: GPS, Ashby (2003); deflection and perihelion, Will (2014); PSR B1913+16, Weisberg & Huang (2016); millimetre-scale redshift, Bothwell et al. (2022); lunar laser ranging, Müller et al. (2023); light deflected by light, Tolman et al. (1931); tides, the S2/M2 equilibrium ratio.

Test	Reading of P	Computed	Measured / known
GPS satellite clock offset	value, with $v^2/2c^2$	$+38.51\,\mu\mathrm{s\,d^{-1}}$	$+38.58\,\mu\mathrm{s\,d^{-1}}$ (built in)
Light deflection, solar limb	$\alpha = 4P$	$1.751''$	$1.75''$
Perihelion advance, Mercury	$6\pi P$ per orbit	$42.98''\,\mathrm{cy^{-1}}$	$43''\,\mathrm{cy^{-1}}$
Orbital decay, PSR B1913+16	Peters–Mathews	$1.0001\times$ GR	$(0.9983 \pm 0.0016)\times$ GR
Redshift across 1 mm	value	1.09×10^{-19}	$(0.98 \pm 0.23) \times 10^{-19}$
Nordtvedt η, Earth–Moon	gradient	0	$(-0.2 \pm 1.1) \times 10^{-4}$
Light deflected by light	$\alpha = 8GE/c^4 b$ head-on	0 side by side	same
Solar/lunar tide ratio	Hessian	0.46	0.46

Table 2: The nested sequence of r_{own} from Eq. (2). The Sun uses $V_c = 230\,\mathrm{km\,s^{-1}}$ at 8.2 kpc; the Milky Way and M31 use Roche lobes for 1.0 and $1.5 \times 10^{12}\,M_\odot$ separated by 780 kpc.

Body	Environment	r_{own}
Moon	Earth (point mass)	61,500 km
Earth	Sun (point mass)	1.50×10^6 km
Sun	Milky Way (flat rotation curve)	1.4 pc
Milky Way / M31	each other (Roche lobes)	269 / 323 kpc
Local Group	empty space, $S = \Lambda c^2/3$	1.38–1.90 Mpc for 1.9–$5 \times 10^{12}\,M_\odot$

- **Extended host** with a flat rotation curve V_c: $S = 2V_c^2/D^2$, which gives the Jacobi tidal radius (King, 1962).
- **Empty space**: the background acceleration $+\frac{1}{3}\Lambda c^2 r$ is isotropic and the same everywhere, so $S = \Lambda c^2/3$. This gives $r_{\mathrm{own}} = R_\Lambda = (3Gm/\Lambda c^2)^{1/3}$, the zero-gravity radius and largest turnaround radius (Pavlidou & Tomaras, 2014).

For comparable masses we use the volume-equivalent Roche-lobe radius (Eggleton, 1983), which for small mass ratios is 0.71 of the L1 distance. Equation (2) contains no new physics. What it adds is a single quantity, S, that characterises the environment of any bound system, with empty space as the final environment.

Thought experiment: pebbles at the edge. Release pebbles at increasing distances from the Moon. Close in they fall back to the Moon; far out Earth's stretch carries them away; in between the two balance. Repeat with Earth in the Sun's stretch, the Sun in the Galaxy's, and finally a galaxy group in empty space, where the stretch is uniform. One procedure yields every boundary in Table 2.

3.2 Checks from planets to clusters

Moons. The outermost known moon of each of six planets (JPL Solar System Dynamics) lies inside r_{own}, and inside the stability limits of about $0.5\,r_{\mathrm{own}}$ for prograde and $0.7\,r_{\mathrm{own}}$ for retrograde orbits (Hamilton & Krivov, 1997). The largest ratios are 0.26 (prograde, the Moon) and 0.46 (retrograde, Jupiter's Kore). Saturn's S/2004 S 52 lies at 0.40, Uranus's Ferdinand at 0.29, Neptune's S/2021 N 1 at 0.44 and Mars's Deimos at 0.02 (Fig. 1a).

Satellite galaxies. Using distances from the Local Volume Database (Pace, 2024), 67 of 68 Milky Way satellites and 40 of 43 M31 satellites lie inside their hosts' r_{own} (Fig. 1b). The four

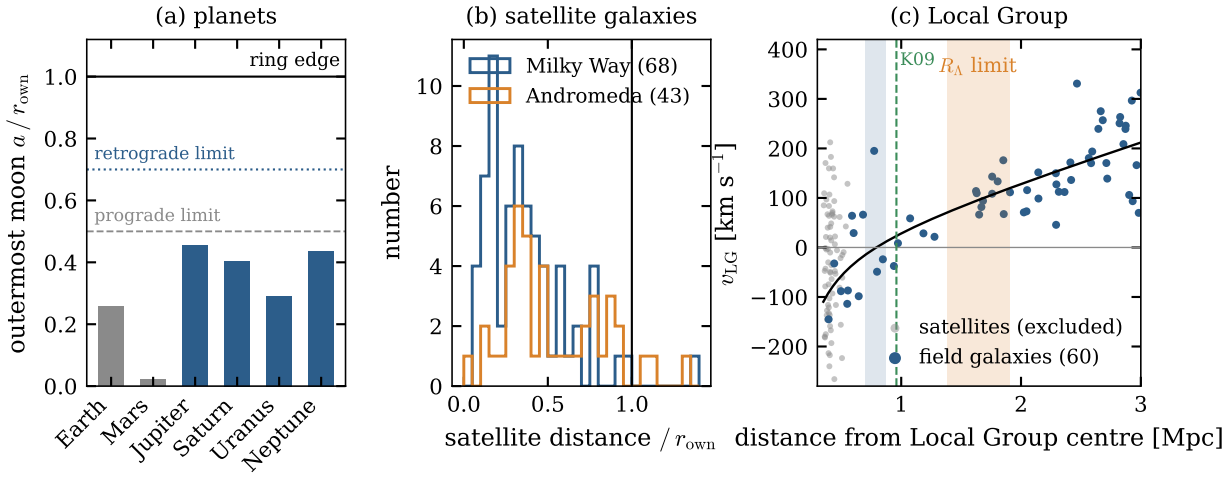


Figure 1: The balance rule $r_{\rm own}^3 = Gm/S$ at three scales. (a) Outermost known moon of six planets as a fraction of $r_{\rm own}$; dashed and dotted lines are the prograde and retrograde stability limits. (b) Distances of 68 Milky Way and 43 M31 satellites as fractions of their hosts' $r_{\rm own}$. (c) Velocities relative to the Local Group centre; blue points are the 60 field galaxies used in the fit (black curve), grey points are excluded satellites. The blue band is the fitted zero-velocity radius, the green dashed line the value of Karachentsev et al. (2009), and the orange band R_Λ for 1.9–$5 \times 10^{12}\,M_\odot$.

outside (Eridanus II, And XVIII, And XXVIII, Perseus I) lie 5–38% beyond. Since $r_{\rm own} \propto M^{1/3}$, a 50% larger host mass moves the edge out by 14%, so these are consistent with sitting at the edge or falling in for the first time.

The Local Group. We take the 132 Local Volume Database galaxies at 0.35–3 Mpc from the Local Group centre and remove the 72 within 300 kpc of the Milky Way or 330 kpc of M31, whose motions are orbital. Fitting the remaining 60 with $v = Hr\,[1 - (R_0/r)^2]$ gives $R_0 = 0.79^{+0.07}_{-0.10}$ Mpc, $H = 76\,{\rm km\,s^{-1}\,Mpc^{-1}}$ and $64\,{\rm km\,s^{-1}}$ rms (Fig. 1c). Galaxy by galaxy, the sign change lies between Leo A (0.94 Mpc, $-37\,{\rm km\,s^{-1}}$) and Aquarius (0.97 Mpc, $+9\,{\rm km\,s^{-1}}$), matching the published $R_0 = 0.96 \pm 0.03$ Mpc (Karachentsev et al., 2009). Both lie well inside $R_\Lambda = 1.38$–1.90 Mpc.

The Virgo cluster. Its zero-velocity radius, 7.2 ± 0.7 Mpc for $(8.0 \pm 2.3) \times 10^{14}\,M_\odot$ (Karachentsev et al., 2014), lies inside $R_\Lambda = 10.3^{+0.9}_{-1.1}$ Mpc. Both groups sit near $0.7\,R_\Lambda$. Because their masses are derived partly from R_0, these two are consistency checks; the moon and satellite checks are not affected by this.

3.3 Empty space as the outermost environment

The a_0 proposal below uses Λ, so we check it independently with 1,590 Pantheon+ supernovae at $z > 0.01$ and their full statistical and systematic covariance (Scolnic et al., 2022; Brout et al., 2022). A flat universe with matter and Λ gives $\Omega_\Lambda = 0.668 \pm 0.018$, in agreement with the Planck value of 0.685 (Planck Collaboration, 2020). Letting the equation of state vary gives $w = -0.95 \pm 0.04$, consistent with a constant ($w = -1$). Universes without Λ fit worse by $\Delta\chi^2 = 48$ (empty, coasting), 155 (open, matter only) and 698 (flat, matter only).

4 The galactic acceleration scale from a presence integral

4.1 Motivation

The RAR is well described by (McGaugh et al., 2016)

$$g_{\rm obs} = \frac{g_{\rm bar}}{1 - e^{-\sqrt{g_{\rm bar}/a_0}}}. \tag{3}$$

In the nested sequence of Section 3, empty space is the outermost environment of every system, and its acceleration scale is

$$a_\Lambda \equiv c^2\sqrt{\Lambda/3} = cH_0\sqrt{\Omega_\Lambda}, \tag{4}$$

the surface gravity of the de Sitter horizon (Gibbons & Hawking, 1977).

Thought experiment: a galaxy alone in empty space. Place one galaxy in a universe containing only Λ. Close in, its own pull dominates. Far out, its pull keeps weakening, while empty space supplies a scale a_Λ that is the same everywhere. The proposal is that the galaxy's dynamics change character where its own pull falls to a_Λ shared over its ring, $a_0 = a_\Lambda/k$, with k fixed by how a point source's presence fills a ring. The thought experiment motivates the form; it does not derive it.

4.2 The factor k from the presence integral

We define k as the total presence of a point source inside a ring of radius R centred on it, divided by its presence at the edge, in units of R^n for an n-dimensional ring:

$$k \equiv \frac{1}{P_{\rm edge}\,R^n}\int_{r<R} P\,dV, \qquad P \propto \frac{1}{r}. \tag{5}$$

With $dV = S_{n-1}r^{n-1}dr$, where S_{n-1} is the area of the unit sphere in n dimensions (Appendix A),

$$k = \frac{S_{n-1}}{n-1} = \begin{cases} 2\pi/1 = 2\pi & \text{planar ring } (n=2), \\ 4\pi/2 = 2\pi & \text{spherical ring } (n=3). \end{cases} \tag{6}$$

The factor π is not inserted by hand: it comes from the ring being round. The planar and spherical results coincide, so the prediction does not depend on whether a galaxy's ring is taken as a disc or a ball. This equality is special to two and three dimensions; for $n = 4$, $k = 2\pi^2/3$. Hence

$$a_0 = \frac{c^2\sqrt{\Lambda/3}}{2\pi} = \frac{cH_0\sqrt{\Omega_\Lambda}}{2\pi} = \begin{cases} 0.863\times10^{-10}\,\mathrm{m\,s^{-2}} & \text{Planck: } H_0 = 67.4,\ \Omega_\Lambda = 0.685, \\ 0.944\times10^{-10}\,\mathrm{m\,s^{-2}} & \text{SH0ES: } H_0 = 73.0,\ \Omega_\Lambda = 0.70, \end{cases} \tag{7}$$

with H_0 in $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ from Planck Collaboration (2020) and Riess et al. (2022). Equation (7) is Milgrom's coincidence (Milgrom, 1999, 2016); what is new is the reason for the 2π. In an earlier version of this work we wrote $k = \pi\langle P\rangle_{\rm in}/P_{\rm edge}$. That form multiplied a two-dimensional factor (π, the area of the unit disc) by a three-dimensional average and gave $3\pi/2$ for a ball. The consistent integral, Eq. (5), removes that inconsistency and the second value.

Three properties follow:

1. **No free parameter.** Given Λ, Eq. (7) has one value.
2. **Universality.** *Thought experiment: the photocopied galaxy.* Build an exact copy of a galaxy ten times larger. Because $P \propto 1/r$ has no scale, Eq. (5) is unchanged, and so is a_0. The same holds for a heavier copy and, since planar and spherical rings give the same k, for a flatter or rounder one. a_0 therefore cannot depend on a galaxy's size, mass or visible shape.
3. **No evolution.** a_0 is tied to Λ, not to the expansion rate, so it is constant in time.

4.3 Data and fitting

We use SPARC (Lelli et al., 2016): 3,391 rotation-curve points in 175 galaxies, of which 2,803 points in 164 galaxies have velocity errors below 10%. For each point $g_{\rm obs} = V_{\rm obs}^2/R$ and $g_{\rm bar} = (V_{\rm gas}^2 + \Upsilon_{\rm disk} V_{\rm disk}^2 + \Upsilon_{\rm bul} V_{\rm bul}^2)/R$, with fiducial $\Upsilon_{\rm disk} = 0.5$ and $\Upsilon_{\rm bul} = 0.7$ at $3.6\,\mu$m (Lelli et al., 2016). When we vary $\Upsilon_{\rm disk}$ we scale the bulge value with it ($\Upsilon_{\rm bul} = 1.4\,\Upsilon_{\rm disk}$). We fit $\log g_{\rm obs}$ with Eq. (3) by

- minimising the rms of log residuals;
- minimising χ^2 with velocity-error uncertainties $\sigma = \max[2(\delta V/V)/\ln 10,\, 0.02]$ dex;
- minimising χ^2 with an added intrinsic scatter $\sigma_{\rm int}$ chosen so that $\chi^2/{\rm dof} = 1$.

Statistical errors come from bootstrapping galaxies. The *gas-rich* subsample contains the 16 galaxies in which gas supplies more than 60% of $g_{\rm bar}$, on average over points with $g_{\rm bar} < 3 \times 10^{-10}\,{\rm m\,s^{-2}}$: D631-7, DDO 154, DDO 161, DDO 168, DDO 170, ESO 444-G084, F565-V2, KK98-251, NGC 3109, NGC 3741, UGC 00731, UGC 00891, UGC 04483, UGC 05764, UGCA 442 and UGCA 444. Per-galaxy fits use galaxies with at least 8 points, excluding fits at the grid boundary (110 galaxies). Disk scale lengths are estimated as $R(V_{\rm disk}^{\rm max})/2.2$. We do not marginalise over distances and inclinations (Li et al., 2018).

4.4 Results at $z \approx 0$

Gas-rich galaxies. These are the cleanest test, because stellar mass-to-light ratios barely matter. The best fit is $a_0 = 0.81 \times 10^{-10}\,{\rm m\,s^{-2}}$, and $0.82^{+0.22}_{-0.17} \times 10^{-10}\,{\rm m\,s^{-2}}$ with a galaxy bootstrap. It changes little with $\Upsilon_{\rm disk}$ (0.92, 0.81 and $0.72 \times 10^{-10}\,{\rm m\,s^{-2}}$ for 0.25, 0.5 and 0.75) and with the gas threshold (0.78, 0.81 and $0.74 \times 10^{-10}\,{\rm m\,s^{-2}}$ for 50%, 60% and 70%, i.e. 25, 16 and 11 galaxies). The prediction is consistent: $\Delta\chi^2 = +0.1$ (Planck) and +2.7 (SH0ES) relative to the best fit, with $\sigma_{\rm int} = 0.170$ dex.

All galaxies. At the fiducial $\Upsilon_{\rm disk} = 0.5$ the rms estimator gives $a_0 = 1.13 \times 10^{-10}\,{\rm m\,s^{-2}}$ (scatter 0.144 dex), the bootstrap $1.14 \pm 0.08 \times 10^{-10}\,{\rm m\,s^{-2}}$, and the intrinsic-scatter likelihood ($\sigma_{\rm int} = 0.134$ dex) $1.14 \times 10^{-10}\,{\rm m\,s^{-2}}$. This is 3.5σ above the Planck prediction on the bootstrap error alone. The full-sample estimate is, however, strongly degenerate with Υ: it falls from 1.47 to $0.90 \times 10^{-10}\,{\rm m\,s^{-2}}$ as $\Upsilon_{\rm disk}$ goes from 0.4 to 0.6 (Fig. 3a), and it matches the prediction at $\Upsilon_{\rm disk} = 0.62$ (Planck) or 0.58 (SH0ES). At $\Upsilon_{\rm disk} = 0.6$ the Planck prediction is within $\Delta\chi^2 = +8.4$ of the best fit. Figure 2 shows that with $\Upsilon_{\rm disk} = 0.62$ the low-acceleration residuals vanish but the highest-acceleration points, where stars dominate, fall about 0.1 dex below, which suggests the inner disks prefer a lower Υ. Points within a galaxy share its distance and inclination errors, so these $\Delta\chi^2$ values overstate significances.

Universality. Per-galaxy values have median $1.15 \times 10^{-10}\,{\rm m\,s^{-2}}$ and scatter 0.26 dex, much larger than their 0.02 dex formal errors, as expected from distance, inclination and Υ errors (Li et al., 2018). They show no correlation (Spearman) with

- disk scale length ($\rho = -0.09$, $p = 0.35$), outermost radius (-0.11, 0.26) and distance (-0.07, 0.49);
- central surface brightness (-0.04, 0.69) and maximum rotation speed ($+0.04$, 0.71);
- gas fraction ($+0.01$, 0.90) and bulge fraction (-0.06, 0.56).

Bulge-rich (bulge > 10% of the baryonic mass) and pure-disc galaxies have median a_0 of 1.07 and $1.15 \times 10^{-10}\,{\rm m\,s^{-2}}$ (Mann–Whitney $p = 0.76$); see Fig. 3b. This is the behaviour required by

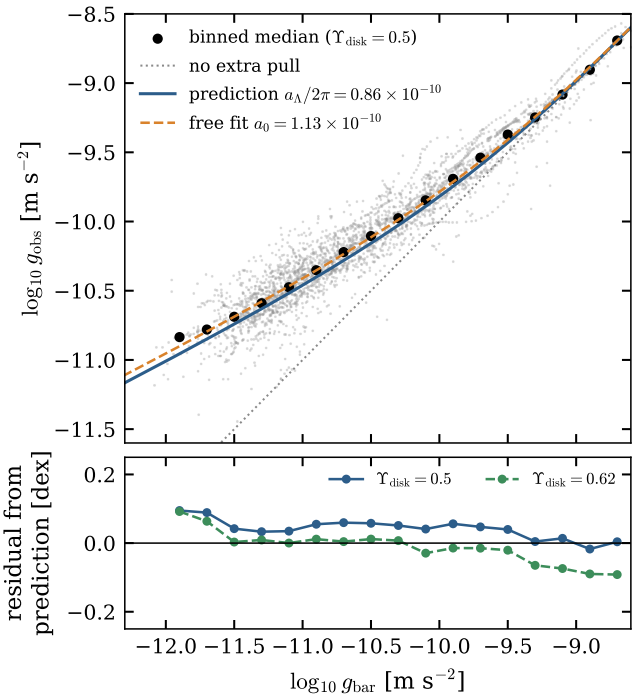


Figure 2: The radial acceleration relation for 2,803 SPARC points in 164 galaxies at $\Upsilon_{\rm disk} = 0.5$, with the predicted $a_0 = a_\Lambda/2\pi$ (Planck) and the free fit. Black points are binned medians. The lower panel shows median residuals from the prediction at $\Upsilon_{\rm disk} = 0.5$ and at 0.62, where the full-sample fit equals the prediction.

property 2: neither size nor visible shape sets a_0. As a contrast, a variant that computes a separate ring factor for each galaxy from its own mass layout predicts $a_0 = 1.29$–$1.71 \times 10^{-10}\,\mathrm{m\,s^{-2}}$. It fits worse ($\chi^2 = 53{,}818$ against 49,907 for one universal value, velocity errors only, 113 galaxies), and the galaxies do not follow its predictions ($\rho = 0.00$, $p = 0.97$). The choice of transition function in Eq. (3) changes a_0 by less than 3% ("simple" function: 1.18 and $0.80 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ for all and gas-rich galaxies, against 1.22 and $0.81 \times 10^{-10}\,\mathrm{m\,s^{-2}}$).

4.5 Does a_0 change with redshift?

Property 3 distinguishes the proposal from relations tied to the expansion rate, $a_0 \propto H(z)$, which would make a_0 1.8 times larger at $z = 1$ and 3 times larger at $z = 2$. We test it with 41 star-forming disks at $z = 0.65$–2.45 whose redshift, disk effective radius R_e, baryonic mass and dark matter fraction $f_{\rm DM}(R_e)$ are tabulated by Price et al. (2021). If the local RAR holds at R_e, then $f_{\rm DM} = \exp(-\sqrt{g_{\rm bar}/a_0})$, so each galaxy gives

$$a_0 = \frac{g_{\rm bar}(R_e)}{[\ln f_{\rm DM}(R_e)]^2}, \tag{8}$$

with $g_{\rm bar}(R_e)$ from half the baryonic mass in a thin exponential disk.

Within the sample, a_0 shows no rise with redshift (Fig. 4; Spearman $\rho = +0.08$, $p = 0.62$). The median for $z > 2$ is 0.89 times that for $z < 1.2$ (68% range 0.78–2.00), where $a_0 \propto H(z)$ predicts 2.06. The fitted slope $d\log a_0/d\log H = +0.36$ (68%: -0.09 to $+0.81$) lies between the two hypotheses, so these data cannot yet exclude evolution. The overall level is 1.5–1.9 times the local value for both disk and spherical mass geometries. Because it is the same at every redshift, we attribute it to calibration (baryonic masses from scaling relations, and model-dependent $f_{\rm DM}$) rather than to evolution, but we cannot exclude it.

5 Discussion

5.1 What is new, and what is not

The presence field, its three readings and the balance rule are reformulations: they reproduce existing physics exactly, and no measurement distinguishes them from the standard description.

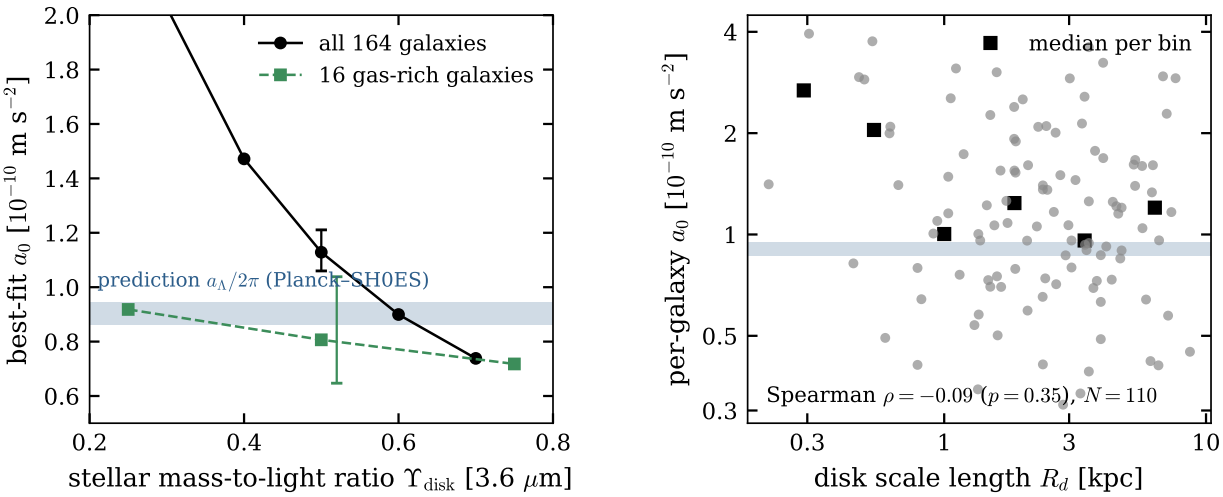


Figure 3: (a) Best-fit a_0 against the assumed stellar mass-to-light ratio, for all 164 galaxies (rms estimator, with bootstrap error at the fiducial value) and for the 16 gas-rich galaxies (χ^2 estimator, with bootstrap error). The band is the prediction $a_\Lambda/2\pi$ for Planck to SH0ES parameters. (b) Per-galaxy a_0 against disk scale length for 110 galaxies, with binned medians and the prediction band.

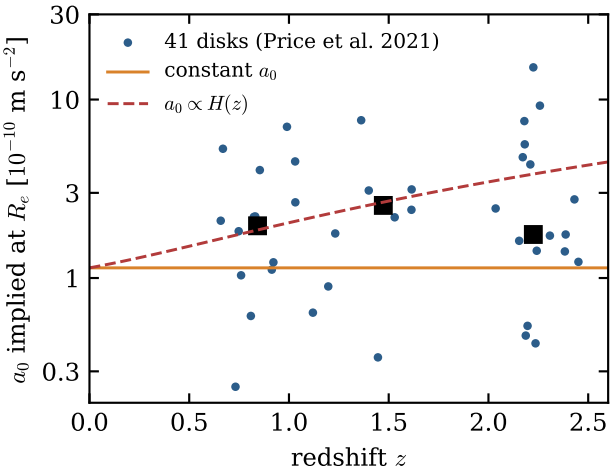


Figure 4: The a_0 implied at the effective radius of 41 disks at $z = 0.65$–2.45 (Price et al., 2021), if the local RAR holds there. Squares are medians in three redshift bins. Lines show a constant a_0 and $a_0 \propto H(z)$, both normalised to the local value. The data show no rise with redshift but a common offset above the local value.

Their contribution is organisational. The one proposal that could change physics is Eq. (7) with its derivation, Eq. (5). It links the acceleration at which galaxy dynamics change character to the emptiness that drives cosmic expansion, with the numerical factor fixed by a geometric integral. If confirmed, it would turn a long-noted coincidence into a relation with a reason. It is not yet confirmed.

5.2 The mass-to-light question and the decisive test

The gas-rich galaxies agree with the prediction; the full sample agrees only if stellar disks are about 20% heavier than the fiducial assumption, which the highest-acceleration points do not favour. The decisive test uses gas-dominated galaxies, for which Υ does not matter, with distances accurate to $\sim 5\%$ (tip of the red giant branch or Cepheids) and well-measured inclinations. In the low-acceleration regime the inferred $a_0 \propto D^{-2} \sin^{-4} i$, so distance and inclination errors dominate the current scatter. Rotation curves of the LITTLE THINGS dwarfs (Iorio et al., 2017), combined with tip-of-the-red-giant-branch distances, are suited to this. A rough baryonic Tully–Fisher estimate for six of them, using catalogue stellar masses, gives 0.84 to 1.12×10^{-10} m s^{-2} depending on those masses, so it does not yet decide. If such a sample reduces the per-galaxy scatter to

about 0.12 dex, roughly twenty galaxies would measure a_0 to about 6%. That is enough to confirm 0.86–$0.94 \times 10^{-10}\,\mathrm{m\,s^{-2}}$, or to exclude it if the true value is near 1.15.

5.3 Relation to modified dynamics and dark matter

Equation (7) fixes the value of a_0 given Λ, but not whether the RAR arises from modified dynamics or from dark matter. It does not by itself explain all dynamics: velocity dispersions of 73 Milky Way and M31 satellites, predicted from their luminosities and the RAR before the measured values were loaded, fall short of the measurements by a median factor of 1.45 for the 35 classical satellites and 3.4 for the 38 ultra-faint ones. We therefore remain agnostic about the origin of the RAR and claim only the value of its scale.

5.4 Limitations

- The presence description is equivalent to general relativity in weak fields and makes no new predictions there.
- Eq. (5) is a definition chosen because it uses presence consistently and gives a shape-independent result. It is not derived from an action, and other plausible definitions would give other factors.
- The relation was formulated after the SPARC value of a_0 and the 2π coincidence were known. It was revised once, from $\pi\langle P\rangle/P_{\mathrm{edge}}$ to Eq. (5), after we found the earlier form mixed dimensions.
- Our fits fix Eq. (3) and do not marginalise over distances, inclinations or per-galaxy Υ. The high-redshift test relies on published dark matter fractions and simple geometry.

The independent tests in Sections 4.5 and 5.2 use data not involved in formulating the proposal.

6 Conclusions

1. Weak-field gravity can be read from one scalar, $P = -\Phi/c^2$: its value gives clock rates, its gradient accelerations and its Hessian tides. This reproduces the standard tests (Table 1).
2. The Hill, Jacobi, L1 and Λ-turnaround radii are one balance condition, $r^3 = Gm/S$, with empty space as the final environment. Moons, satellite galaxies, the Local Group and Virgo all sit inside the boundaries it gives, and 1,590 supernovae independently confirm the Λ it uses.
3. The total presence of a point source inside a ring, relative to its edge value, equals 2π for both planar and spherical rings. This gives a parameter-free prediction, $a_0 = c^2\sqrt{\Lambda/3}/2\pi = 0.86$–$0.94 \times 10^{-10}\,\mathrm{m\,s^{-2}}$, and a reason for the known 2π coincidence.
4. Gas-rich galaxies agree with it; the full sample agrees if $\Upsilon_{\mathrm{disk}} \approx 0.6$. a_0 is independent of galaxy size, distance, brightness, speed, gas fraction and visible shape, and shows no rise with redshift up to $z \approx 2.4$.
5. Gas-dominated galaxies with precise distances can confirm or exclude the prediction.

Data and code availability

All numbers and figures come from `paper_analysis.py`, `paper_additions.py` and the earlier test scripts in the public repository https://github.com/hassan-algo/rings-of-presence. The repository includes the SPARC rotation curves (CC BY 4.0; Lelli et al. 2016) and the Local Volume Database

tables (CC0; Pace 2024); the Pantheon+ data are downloaded by a script. The high-redshift values are from Table 3 of Price et al. (2021). Moon orbits are from the JPL Solar System Dynamics planetary satellite mean elements.

Acknowledgements

I thank the SPARC team, A. B. Pace for the Local Volume Database, the Pantheon+ collaboration and the JPL Solar System Dynamics group for making their data public. *Use of AI tools:* I used Claude (Anthropic), a large language model, as a research assistant. It helped turn my ideas into equations, wrote and ran the analysis code against the public data, and helped draft the text. Every result was generated by the public scripts and checked against the cited measurements. I take full responsibility for the content of this paper.

A The presence integral in n dimensions

For $P \propto 1/r$ about the centre of an n-dimensional ball of radius R, with $dV = S_{n-1} r^{n-1} dr$ and $S_{n-1} = 2\pi^{n/2}/\Gamma(n/2)$,

$$\frac{1}{P_{\rm edge} R^n} \int_0^R \frac{1}{r}\, S_{n-1} r^{n-1}\, dr = \frac{R}{R^n}\, S_{n-1} \frac{R^{n-1}}{n-1} = \frac{S_{n-1}}{n-1}. \tag{9}$$

With $S_1 = 2\pi$ and $S_2 = 4\pi$ this gives 2π for $n = 2$ and $n = 3$, independent of R; $S_3 = 2\pi^2$ gives $2\pi^2/3$ for $n = 4$.

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